

CADA Marketing HQ

System Architecture

Technical Architecture & Feature Documentation

June 2026

cada-marketing-hq.vercel.app

1. Overview

CADA Marketing HQ is an AI-powered marketing platform built exclusively for **CADA**, a modest fashion brand serving Muslim women in Indonesia and Singapore. It centralises content creation, trend research, campaign planning, social publishing, and performance analysis into a single workspace — powered by a suite of leading AI models.

Live URL	cada-marketing-hq.vercel.app
Repository	github.com/juliautomato/CADA-marketing-hq
Platform	Vercel (auto-deploy on push to main)
Target	CADA brand team — content creators, founders, marketing managers
Markets	Indonesia, Singapore — Muslim women aged 20–35

2. Technology Stack

Framework	Next.js 16 (App Router, TypeScript)
Styling	Tailwind CSS + Framer Motion + custom components
Database	Supabase Postgres (project: pyuvofppbfuytkcazgwh, region ap-northeast-1)
File Storage	Supabase Storage — buckets: product-images, videos
Hosting	Vercel — cada-marketing-hq.vercel.app
Auth/Oauth	Instagram Graph API OAuth, TikTok OAuth 2.0

3. AI Providers & Logic

Seven AI models handle distinct specialities within the platform:

Provider	Model	Role
Anthropic	Claude Sonnet 4.6	All text generation — captions, descriptions, emails, trend research, campaign planning, pe
Anthropic	Claude Haiku 4.5	Lightweight tasks — prompt enhancement, brand website analysis (fast & cost-efficient)
OpenAI	GPT Image 1	Image generation with reference editing (images.edit). Sizes: 1024×1024, 1024×1536. Qua
fal.ai	Flux 1.1 Pro	Photorealistic image generation. Aspect ratios: 1:1, 9:16. Default provider for AI Image task
Fashn.ai	tryon-v1.6	Virtual Try-On — garment photo + model photo → model wearing exact garment. Polls /v1/
Runway	Gen4.5 + Seedance2	Text-to-video and image-to-video. Gen4.5 for landscape; Seedance2 (runway-ref) for multi-
Kling AI	kling-v2-master	Video generation, 9:16 portrait default. JWT authentication.

4. Pages & Features

Dashboard

[/](#)

Live stats (runs, content items, scheduled posts), recent agent runs, quick-action cards.

Content Creator

[/agents/creator](#)

8 content types: Caption, Description, Email, AI Image (Flux/GPT), Short Video (Kling/Runway/Seedance2), IG Story (image or video), Virtual Try-On (Fashn.ai), Canva Template. Features: catalog picker, reference image upload, AI prompt enhancer (Haiku), image provider toggle, download buttons, post to Instagram/TikTok, schedule post, content library with pagination.

Trend Analyst

[/agents/trend](#)

Live web search + Pexels mood board + hashtag & creator tracking.

Campaign Planner

[/agents/campaign](#)

4-week calendar; exports tasks to Todoist, events to Google Calendar, brief to Drive.

Performance Reviewer

[/agents/performance](#)

Live Instagram metrics via Graph API (reach, plays, likes, saves, comments). AI Analyse tab sends data to Claude for insights.

Full Campaign Agent

[/agents/full-campaign](#)

8-step multi-agent chain — brief, research, creative, copy, schedule, assets, report. Streams via SSE.

Post Scheduler

[/scheduler](#)

Queue posts with date + time slot. 5 publish windows: 9am/12pm/3pm/6pm/9pm WIB. Per-post "Publish Now" button.

Product Catalog

[/products](#)

CRUD for garments — name, description, price, image upload to Supabase Storage.

Automations

[/automations](#)

Enable/disable scheduled agents, manual trigger button, last run timestamp.

History

[/history](#)

Paginated log of all agent runs with status, duration, input/output.

Settings — Brand

[/settings](#)

Voice & tone, content guidelines, target customer, campaign theme, caption examples, AI image quality selector. Auto-fill from website URL (Haiku analysis).

Settings — Connections

[/settings](#)

Instagram OAuth, TikTok OAuth, Google Drive media folder ID + upload toggle.

5. How Logic Flows — Key User Journeys

Journey 1 — Generate & Post an AI Image

- User selects AI Image task in Content Creator
- Picks provider: Flux 1.1 Pro (photorealistic) or GPT Image 1 (reference editing)
- Types a rough scene description in the textarea
- Optionally uploads a reference photo (stored in Supabase Storage)
- Clicks "Enhance Prompt" → API calls Claude Haiku → rewrites description into detailed photography prompt (preserving user's original instructions)
- Clicks Generate → POST /api/agents/creator with { task: "image", imageProvider, prompt, referenceImageUrl }
- API calls Flux (fal.ai fal-ai/flux-pro/v1.1) or GPT Image 1 (images.generate / images.edit)
- Image URL / base64 returned → optionally auto-uploaded to Google Drive folder
- Saved to cada_content_items table with image_url, drive_url, metadata
- Displayed with Download button, Post to Instagram, Schedule buttons

Journey 2 — Virtual Try-On

- User uploads garment photo + model photo (both stored in Supabase Storage)
- Clicks Generate → POST /api/agents/creator with { task: "tryon", referenceImageUrl (garment), referenceImageUrls[0] (model) }
- API calls POST https://api.fashn.ai/v1/run with model_name: tryon-v1.6
- Polls GET https://api.fashn.ai/v1/status/{id} every 3 seconds (up to 2 minutes)
- On status "completed" → result image URL returned
- Saved to cada_content_items as type "tryon"
- Downloadable or posted directly to Instagram

Journey 3 — Post Scheduling

- User clicks Schedule on any generated content
- Picks platform (Instagram / TikTok) and time slot (9am / 12pm / 3pm / 6pm / 9pm WIB)
- POST /api/scheduled-posts → saved to cada_scheduled_posts table (status: pending)
- Vercel cron runs 5× daily at UTC 02:00 / 05:00 / 08:00 / 11:00 / 14:00
- publish-posts route queries posts WHERE scheduled_at <= now AND status = pending
- Calls Instagram Graph API (mediaType=IMAGE/REELS/STORIES) or TikTok Content Posting API
- Updates row status to published (with post_id) or failed (with error)

Journey 4 — Brand-Aware Content Generation

- Every content generation request calls getSystemPrompt(db)
- Reads brand_voice, brand_guidelines, brand_target_customer, brand_campaign_theme, brand_caption_examples from cada_settings

- Values injected into Claude's system prompt via getBrandSystemPrompt()
- All generated text — captions, emails, descriptions, trend reports — reflects CADA's brand identity automatically
- Brand settings editable in Settings → Brand tab; auto-fill from website URL uses Claude Haiku

Journey 5 — Instagram Story

- User selects IG Story task, then picks Image or Video format
- Claude Haiku generates a short 1–2 line punchy text overlay (no hashtags)
- Image: Flux 9:16 (fal.ai) or GPT Image 1 1024×1536
- Video: Kling (portrait default) or Runway Gen4.5 at 720:1280
- Media displayed in portrait preview frame
- Post as Instagram Story → calls /api/instagram/post with mediaType=STORIES via Graph API v25.0

6. Integrations Map

Service	API	Usage
Instagram	Graph API v25.0	Post IMAGE / REELS / STORIES; fetch media insights (reach, plays, likes, saves, c
TikTok	Content Posting API	Draft or direct video post (sandbox in dev mode; requires app review for production)
Google Drive	Drive API v3	Auto-upload generated media; export campaign briefs and performance reports
Google Calendar	Calendar API v3	Create campaign milestone events from Campaign Planner
Todoist	REST API v2	Create campaign tasks from Campaign Planner
Canva	Connect API	Create design templates (designType: INSTAGRAM_POST)
Pexels	Pexels API	Mood board images for Trend Analyst reports
Supabase	JS Client + Storage	Postgres database (service role) + file storage for uploads

7. Database Schema

All tables live in Supabase Postgres (project: pyuvofppbfuytkcazgwh, region ap-northeast-1).

Table	Purpose	Key Columns
cada_agent_runs	Log of every AI agent execution	id, agent, status, input (json), output (json), error, duration_ms
cada_content_items	All generated content	id, type (caption/image/video/story/tryon...), title, body, image_url, video_url, dr
cada_scheduled_posts	Post scheduling queue	id, platform, media_url, caption, scheduled_at, status (pending/published/failed)
cada_posts	Published post tracking	id, platform, post_id, media_url, caption, published_at
cada_settings	Key-value store for all settings	key, value — brand_voice, image_quality, instagram_user_token, drive_media
cada_products	Product catalog	id, name, description, price, image_url, created_at
cada_campaigns	Campaign records	id, title, objective, dates, status
cada_campaign_milestones	Per-week campaign tasks	id, campaign_id, week, title, tasks (json)
cada_trend_reports	Trend research outputs	id, title, content, mood_board_urls, created_at
cada_performance_reports	Performance analysis	id, period, metrics (json), insights, created_at

cada_settings — Key Groups

- **Brand:** brand_voice, brand_guidelines, brand_target_customer, brand_campaign_theme, brand_caption_examples, image_quality
- **Instagram:** instagram_user_token, instagram_page_token, instagram_page_id, instagram_business_account_id, instagram_username
- **TikTok:** tiktok_access_token, tiktok_open_id, tiktok_refresh_token, tiktok_username, tiktok_post_mode
- **Google Drive:** drive_media_folder_id, drive_media_upload_enabled

8. Agentic Roadmap

The platform follows a four-level autonomy ladder. Each level builds on the previous, moving from manual-trigger tools toward a fully autonomous AI Marketing Manager.

Level 1	Scheduled Agents (Live)	LIVE
• Monday Trend Brief — Vercel cron 0 1 * * 1 (8am WIB). Runs Trend Analyst, saves report, optionally emails summary. • Daily Content Queue — Vercel cron 0 2 * * * (9am WIB). Generates caption + image for featured product of the day. • Post Publisher (x5) — cron at UTC 2/5/8/11/14 (9am/12pm/3pm/6pm/9pm WIB). Publishes pending scheduled posts.		
Level 2	Trigger-Based Agents (Planned)	PLANNED
• New Product Agent — fires when a product is added to the catalog. Auto-generates caption + AI image + Shopee description. • Low Engagement Alert — fires when a post's 24h reach falls below threshold. Suggests a follow-up Story or boosted post.		
Level 3	Multi-Step Agents	LIVE / PLANNED
• Full Campaign Agent (LIVE) — 8-step chain: brief intake, trend research, creative direction, copy, asset generation, scheduling, calendar export, performance baseline. Streams via SSE. • Performance → Content Loop (PLANNED) — weekly: fetch last 7 days metrics, identify top performers, generate more content in that style, auto-queue for next week.		
Level 4	AI Marketing Manager (Future)	FUTURE
• Autonomous 24/7 agent that monitors Instagram performance, detects opportunities (trending audio, viral formats), briefs and generates content, schedules it, and reports back. • Human in the loop: approves final content before publish; can override at any step.		

9. Environment Variables

All secrets are stored in Vercel environment variables. Never committed to the repository.

Supabase

NEXT_PUBLIC_SUPABASE_URL | NEXT_PUBLIC_SUPABASE_ANON_KEY |
SUPABASE_SERVICE_ROLE_KEY

AI — Text

ANTHROPIC_API_KEY

AI — Images

OPENAI_API_KEY | FAL_API_KEY | FASHN_API_KEY

AI — Video

RUNWAYML_API_KEY | KLING_ACCESS_KEY | KLING_SECRET_KEY

Social

META_APP_ID | META_APP_SECRET | TIKTOK_CLIENT_KEY | TIKTOK_CLIENT_SECRET

Google

GOOGLE_CLIENT_ID | GOOGLE_CLIENT_SECRET | GOOGLE_REFRESH_TOKEN |
GOOGLE_CALENDAR_ID

Other

CANVA_CLIENT_ID | CANVA_CLIENT_SECRET | TODOIST_API_KEY | PEXELS_API_KEY |
NEXT_PUBLIC_APP_URL

10. Known Limitations & Notes

GPT Image 1 aspect ratio

True 9:16 is not supported. Stories use 1024×1536 (2:3 ratio) — slight crop on Instagram. Flux 1.1 Pro natively supports 9:16.

GPT Image 1 reference fidelity

`images.edit()` adapts style from a reference but cannot replicate exact garment stitching or cut. Use Virtual Try-On (Fashn.ai) for exact product replication.

Instagram token expiry

Token expires ~60 days in Meta developer mode. Requires Meta app review for a permanent long-lived token.

TikTok posting

Direct Post requires TikTok app review. Currently creates draft posts in sandbox mode.

Google OAuth refresh

Refresh token expires every 7 days in Google OAuth test mode. Requires publishing the app in Google Cloud Console for persistent tokens.

Fashn.ai model gallery

No built-in model gallery — user must upload a model photo manually each time.

Vercel free tier

Function timeout is 10 seconds on hobby plan; video generation (Kling, Runway) can take 60–120 s. Pro plan or background jobs needed for reliability.

CADA

Marketing HQ

Built with Next.js 16 · Supabase · Vercel · Anthropic Claude · OpenAI · fal.ai · Fashn.ai · Runway · Kling AI

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